assignment15 death of massive stars

This is a preview of the draft version of the quiz

Started: Mar 23 at 7:07pm

Quiz Instructions

Select wisely.

Question 1 6 pts

as a massive star collapses, the core

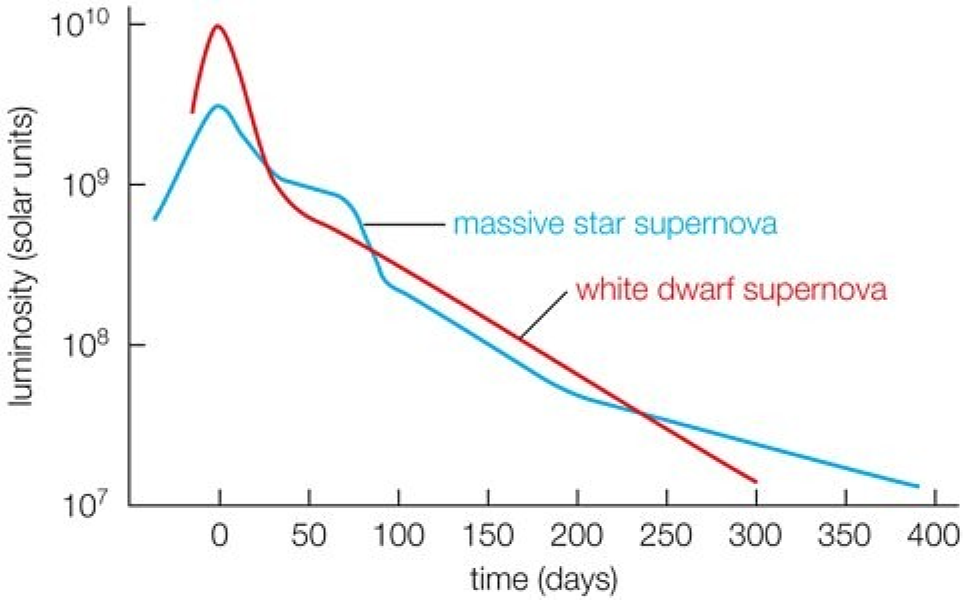
- spins slower
- spins faster
- spins at the same rate
- whaaaat??

Question 2 6 pts

A supernova type IA happens when

- when a red dward collapses
- when a brown dward exploses
- When a white dwarf reaches 1,4 Msun
- a high mass star dies

Question 3 6 pts



The figure above shows how the luminosity of supernovae changes over time. Which type of supernova is the most luminous one year after the explosion?

- white dwarf supernova
- impossible to tell from the graph
- massive star supernova

Question 4 6 pts

Who discovered the limit for the mass of a white dwarf/neutron star (1.4 mass of the Sun)

- Fritz Zwicky
- Albert Einstein
- Dame Jocelyn Bell
- Subrahmanyan Chandrasekhar

Question 5 4 pts

Consider a main-sequence star, a white dwarf, a neutron star, and the singularity of a black hole. Which of the following lists them in the correct order of increasing density (low to high)?

- main sequence star, neutron star, white dwarf, black hole singularity
- black hole singularity, main sequence star, white dwarf, neutron star

main sequence star, black hole singularity, neutron star, white dwarf

main sequence star, white dwarf, neutron star, black hole singularity

Question 6 6 pts

When there is a supernova type II happens

nothing is left hebind

only a neutron star is the remnant

only a BH is the remnant

a BH or a neutron star can be the remnant

Question 7 6 pts

If you were to come back to our Solar System in 8 billion years, what might you expect to find?

(so after the Sun is done using He in C, O

a red giant star

a white dwarf

a black hole

a rapidly spinning pulsar

Everything will be pretty much the same as it is now.

Question 8 6 pts

When discussing stellar lives, astronomers divide stars by mass into low-mass stars (masses similar to the Sun or lower) and high-mass stars (masses more than about 8 times that of our Sun). Which category includes stars that die in supernova explosions?
hint: not including binary systems with white dwarf and red giant

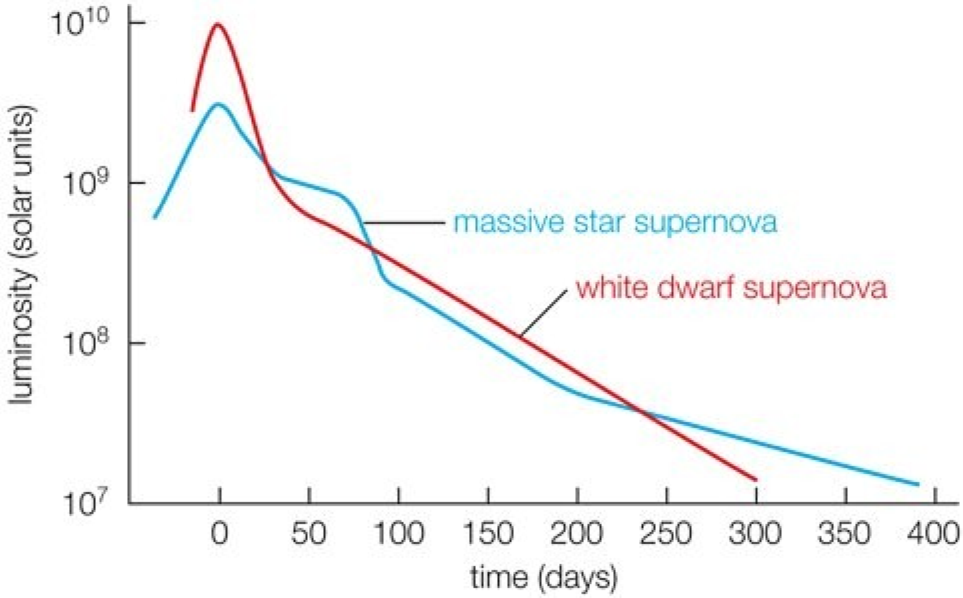
low-mass stars only

high-mass stars only

They both die in supernova explosions.

○
Neither: Supernovas only occur in binary star systems.

Question 9 6 pts



The figure above shows how the luminosity of supernovae changes over time. About how long does it take a white dwarf supernova to decrease in luminosity by a factor of 100 from its peak?
hint: THe y-axis has a log scale. each step means an increase by a factor of 10 (x 10). between 10^7 and 10^8 is a factor of 10.

○
200 days

○
300 days

○
25 days

○
100 days

Question 10 6 pts

WHo predicted pulsars/neutron stars and dark matter? (he came up with the word supernova and he used to call his colleagues sperical b*)

○
Subrahmanyan Chandrasekhar

○
Same Jocelyn bell

○
Albert Einstein

Fritz Zwicky

Question 11 6 pts

Ophiuchus is represented by

a hunter with his bow and arrow

a scorpion with stingers

a vain queen

a man grabbing a snake

Question 12 6 pts

Degeneracy pressure is the source of the pressure that stops the crush of gravity in all the following *except*

the core of a high-mass main-sequence star.

a neutron star.

a white dwarf.

a brown dwarf.

Question 13 6 pts

What would happen if Jupiter was suddenly replaced by a black hole with the same mass as Jupiter?

hint: what matter is the mass . if the mass does not change ...nothing changes

The other planets and the Sun would slowly be pulled into Jupiter.

The other planets would slowly be pulled into Jupiter, but the Sun would be unaffected.

The entire solar system would rapidly be sucked into the black hole.

The orbits of the planets in our solar system would be unaffected.

Question 14 6 pts

Consider our Moon, a white dwarf, and a neutron star. Which of the following lists them in the correct order of size (diameter), from smallest to largest?

hint: slide 4. neutron stars

neutron star, Moon, white dwarf

Moon, white dwarf, neutron star

white dwarf, neutron star, Moon

Question 15 6 pts

Consider a main-sequence star, a white dwarf, a neutron star, and the singularity of a black hole. Which of the following lists them in the correct order of increasing density (low to high)?

main sequence star, black hole singularity, neutron star, white dwarf

main sequence star, neutron star, white dwarf, black hole singularity

main sequence star, white dwarf, neutron star, black hole singularity

black hole singularity, main sequence star, white dwarf, neutron star

Question 16 4 pts

When a massive star collaspes to become a neutron star

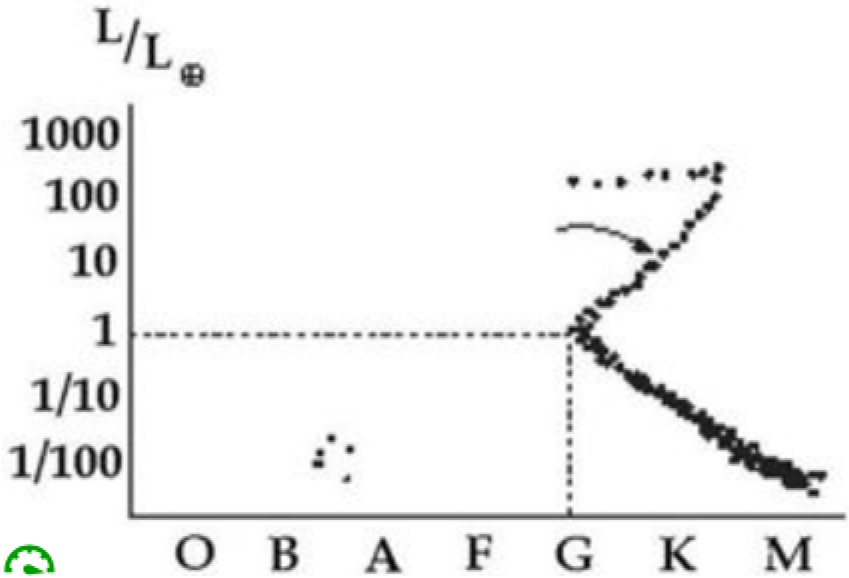
protons turn to neutrons

electrons and protons are ejected to leave behing neutrons

electrons turn to protons

electrons combine with protons to become neutrons

Question 17 4 pts



Refer to the sketch above of an H-R diagram for a star cluster. What is the age of the cluster?
hint: the stars larger than our sun have moved away from the main sequence to become red giants

our sun is a G star and you know how long is life is

- Less than 1 billion years
- About 2 billion years
- About 1 billion years
- About 10 billion years
- More than 15 billion years

Question 18 4 pts

Who discovered the first pulsar

- Albert Einstein
- Fritz Zwicky
- Lord Eddington

○ Dame Jocelyn Bell

Question 19 4 pts

A supernova type II happens

○ when a high mass star dies

○ when a low mass star dies

○ when 2 black holes merge

○ When a white dwarf reaches 1,4 Msun

Question 20 4 pts

Who detect the supernova of 1604?

○ Johannes Kepler

○ William Hershel

○ Galileo Galilei

○ Tycho Brahe

Question 21 4 pts

When the first pulsar was detected, it was called the

○ little green men

○ I missed the class

○ the pulsating neutron

○ big red woem

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